

ST-ETIENNE BOUTHEON AP, France

WMO: 074750

Lat: 45.5331N

Lon: 4.2933E

Elev: 404

StdP: 96.57

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.9	-5.0	-10.5	1.6	-5.1	-8.2	2.0	-3.2	9.4	9.7	8.4	8.7	2.0	330	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	33.1	19.6	31.0	19.2	29.0	18.6	21.0	29.8	20.1	28.5	19.3	27.2	4.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	13.7	23.9	17.4	13.0	23.0	16.6	12.4	22.3	62.4	30.0	59.3	28.6	56.6	27.3	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.9	7.5	6.4	-10.2	36.1	2.8	2.0	-12.2	37.5	-13.8	38.6	-15.4	39.7	-17.4	41.2		
			-10.8	22.7	2.7	0.9	-12.8	23.3	-14.4	23.9	-15.9	24.4	-17.9	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	3.4	4.2	7.6	10.7	14.4	18.6	20.5	20.0	16.5	12.8	7.5	4.4
	DBStd	7.24	4.31	4.51	3.94	3.56	3.82	3.97	3.55	3.61	3.54	4.10	4.21	4.35
	HDD10.0	805	207	166	94	34	6	0	0	0	1	21	97	179
	HDD18.3	2647	462	397	333	230	131	44	18	23	76	175	325	433
	CDD10.0	1447	3	3	19	54	143	259	326	309	197	107	22	4
	CDD18.3	247	0	0	0	1	10	53	85	74	21	3	0	0
	CDH23.3	2759	0	0	1	18	131	603	936	822	229	19	0	0
	CDH26.7	1023	0	0	0	1	28	227	371	342	54	1	0	0
Wind	WSAvg	2.6	2.6	2.8	2.8	2.9	2.7	2.6	2.4	2.3	2.3	2.6	2.7	2.7
Precipitation	PrecAvg	745	42	36	33	54	77	75	75	69	74	81	84	45
	PrecMax	963	94	69	105	139	134	183	177	171	192	161	212	167
	PrecMin	514	5	8	0	5	17	24	26	25	15	10	36	10
	PrecStd	114	22	17	23	31	32	32	33	31	42	38	43	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.7	17.1	21.7	25.8	29.8	34.9	35.6	36.2	30.7	25.8	19.8	15.9
		MCWB	10.7	9.7	11.7	14.6	17.9	20.0	19.9	19.6	18.4	16.4	13.8	11.0
	2%	DB	13.4	14.5	18.9	22.7	26.8	31.7	32.8	33.1	28.3	23.2	17.4	14.1
		MCWB	9.4	9.0	10.7	13.7	16.8	19.6	19.8	19.6	17.8	15.8	12.5	9.9
	5%	DB	11.6	12.7	16.7	20.4	24.3	29.2	30.6	30.6	26.1	21.2	15.8	12.9
		MCWB	8.2	8.1	10.2	12.3	15.7	18.9	19.3	19.1	17.1	15.4	11.6	9.2
	10%	DB	10.0	11.1	14.6	18.1	22.1	27.0	28.5	28.1	23.9	19.6	14.2	11.2
		MCWB	7.1	7.1	9.3	11.5	14.7	18.1	18.7	18.5	16.4	14.5	10.7	8.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	10.6	13.1	15.8	19.5	22.1	22.0	21.7	20.2	17.9	14.2	11.6
		MCDB	14.8	15.2	19.3	24.1	26.6	31.1	31.0	30.7	27.3	23.6	18.7	15.2
	2%	WB	9.7	9.4	11.6	14.2	17.7	20.7	20.9	20.6	18.7	16.7	12.9	10.3
		MCDB	13.1	13.7	17.1	21.0	24.9	29.0	29.7	29.7	26.1	21.7	16.5	13.5
	5%	WB	8.5	8.4	10.6	13.0	16.5	19.6	20.0	19.7	17.7	15.7	12.0	9.5
		MCDB	11.4	12.3	15.9	19.3	22.6	27.1	28.7	28.3	24.1	20.3	15.4	12.4
	10%	WB	7.3	7.4	9.6	11.8	15.3	18.6	19.2	18.9	16.9	14.7	10.9	8.4
		MCDB	9.8	10.7	14.3	17.2	20.8	25.5	27.0	26.7	22.8	18.8	14.0	11.0
Mean Daily Temperature Range	5% DB	MDBR	6.9	8.3	10.4	11.2	11.3	12.2	12.6	12.7	11.6	9.6	7.6	7.0
		MCDBR	9.0	12.0	14.1	15.6	14.8	15.9	16.1	17.0	15.3	12.1	10.3	9.1
		MCWBR	6.0	7.6	8.0	8.1	7.0	6.8	6.4	6.6	6.8	6.4	6.4	5.9
	5% WB	MCDBR	8.6	11.0	12.7	14.5	13.6	14.6	14.8	15.1	13.1	11.1	9.4	8.5
		MCWBR	5.8	7.2	7.3	7.8	6.9	6.7	6.4	6.6	6.3	6.1	6.0	5.7
Clear-Sky Solar Irradiance	taub	0.301	0.322	0.351	0.388	0.393	0.396	0.392	0.387	0.366	0.354	0.320	0.298	
	taud	2.485	2.425	2.364	2.267	2.304	2.329	2.336	2.359	2.398	2.434	2.481	2.493	
	Ebn at Noon	814	859	877	870	875	872	870	862	851	808	779	777	
	Edh at Noon	68	86	105	127	126	124	121	114	101	85	68	61	
All-Sky Solar Radiation	RadAvg	1.28	2.13	3.41	4.50	5.25	6.25	6.16	5.33	4.08	2.57	1.44	1.08	
	RadStd	0.15	0.30	0.41	0.57	0.50	0.50	0.52	0.40	0.31	0.26	0.17	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page